

Molecular Biology Lab

Genotyping of SNPs and mutation

① Nucleic acid Isolation

(DNA , RNA , Plasmid)

- Bacteria.
- Blood cells.

② Quantification and nucleic acid analysis (agarose gel electrophoresis)

③ PCR [==] not necessary gene, may be part of it or exon.

④ Genotyping

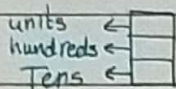
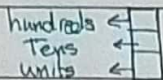
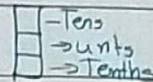
- Allele specific PCR
- RFLP (restriction fragment length polymorphism)
- sequencing
- cloning

Micropipette

Blue pipette

yellow

Red

100-1000 μ L20-200 μ L0.5-10 μ L

DNA Isolation From human blood cells (Promega Kit)

salting out

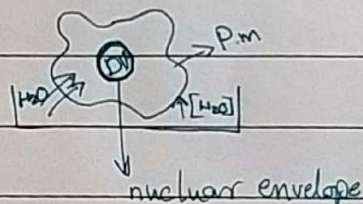
1-cell lysis (TE + salts)

(hypotonic solution)

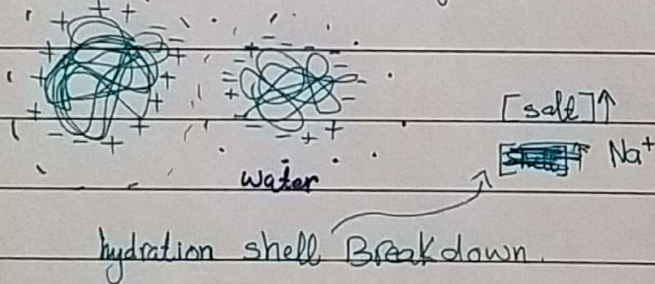
2- Nuclei lysis (TE + salts)

(hypotonic solution)

for nucleases



3- Protein Precipitation (hyper tonic solution)

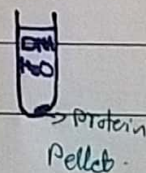


4- DNA precipitation

5- DNA washup

6- Airdrying

7- Rehydration (TE)



RNA isolation

RNAs degrade RNA

RNA isolation is difficult than DNA isolation because the inactivation of RNAs is difficult to

Types of RNA:

- mRNA 2.5-5%

- tRNA

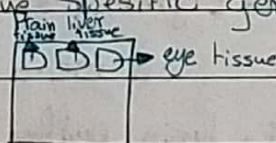
- rRNA > 90% of total RNA in the cells.

why we need to isolate RNA?

1- sequence of mRNA estimation of amino acid sequence.

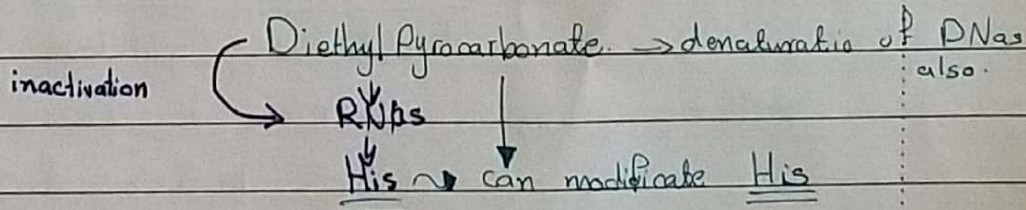
2- length: (Alternative splicing)

3- Quantity: pattern of expression → tissue specific genes
developmental genes

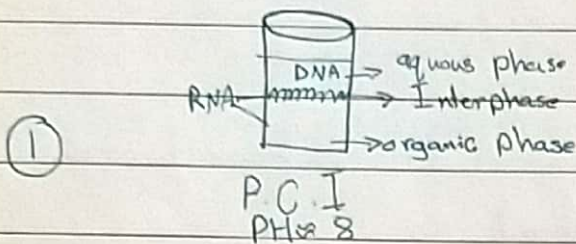


RNA isolation using organic method:

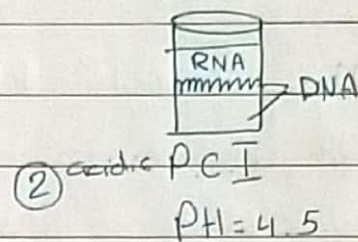
* all solutions or chemical reagents used during RNA isolation must be treated with 1% DEPC.



DNA isolation



RNA isolation



optimum pH of
RNA is 4.5-5.

③ TE → rehydration

RNAs free water
acidic.

RNA isolation using (Absorption column) Qigen Kit.

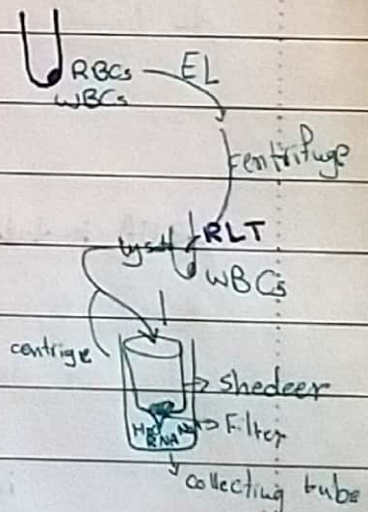
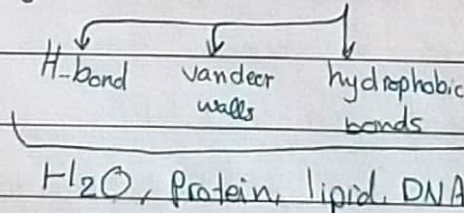
1- Harvesting of cells.

2- Erythrocytes lysis (EL)

on ice → degradation slowly.

3- lysis buffer (RLT)

* Guanidine thiocyanate (chaotropic salt)



* B-mercaptoethanol → Protein degradation

break down for disulfide bond

* salts (Na⁺)

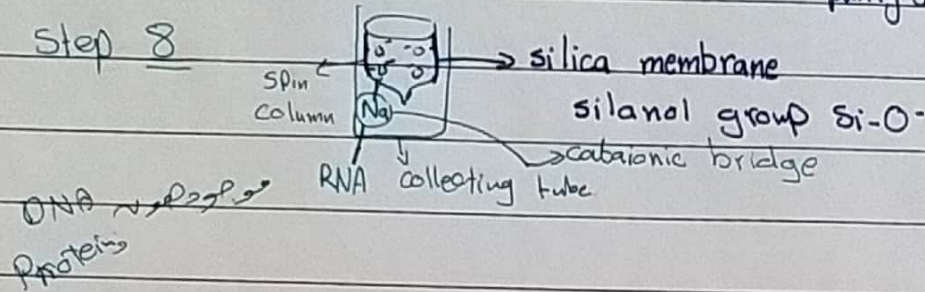
Step 7

ethanol
 → RNA, Na, H₂O, DNA, Protein.

fragmented

purity di 1.26

Step 8



4- spin column.

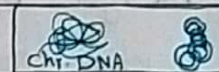
5- washing buffer (RWI RPF) → Guanidine thiocyanate → degradation of DNA and protein.
 I so propanol
 ↓
 H₂O di 1.26

6- elution buffer (RNase free water) dissolve for Na ion

Plasmid isolation

Plasmid: extra chromosomal DNA (Replicon)
can replicate itself independently

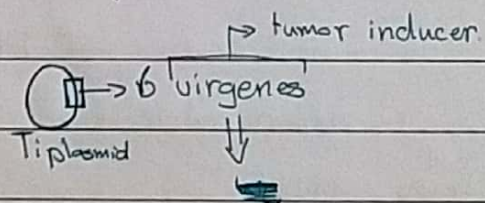
around 300 genes are in the plasmid
genes in the plasmid are not required to growth.
* not in all bacteria.



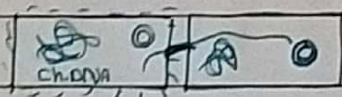
Types of plasmid :

- R plasmid - resistance for antibiotic.
- TI plasmid type from virulence plasmid.

tumor inducer.



- F - plasmid



- HFr- plasmid high frequency recombinant
type of F-plasmid.

- can make integration with chromosomal DNA AFTER conjugation.

- col-plasmid (colchsin)
Kill other type of bacteria

● Degradative plasmid.

degradation of specific type of material

E-coli have a plasmid.

① cell harvesting

② resuspension buffer (TE)

③ cell lysis buffer

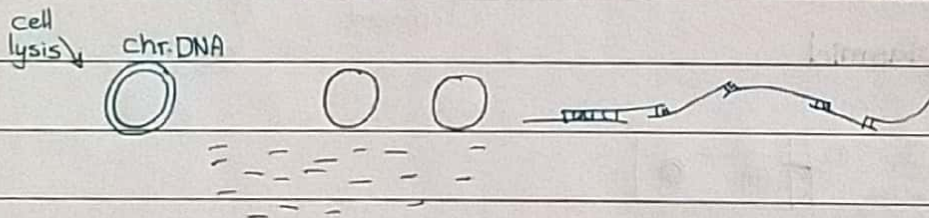
* 10% SDS -ve charged. } denaturation for protein, lipid.
* NaOH PH-12-12.5 } chrDNA, RNA, Plasmid.

④ Neutralization buffer (Potassium Acetate)

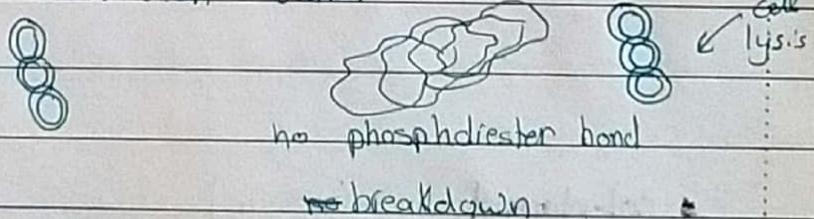
PH 4.5-5

- neutralization of PH around 8.

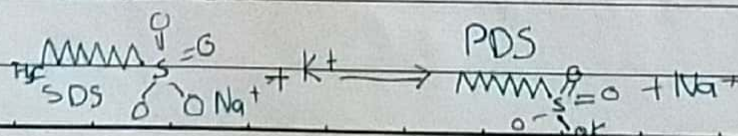
after neutralization chrDNA fail to renatured.



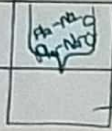
Plasmid can renature because it's supercoiling.
when we put neutralization buffer



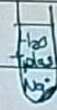
- to ~~set~~ remove SDS.



⑤



→ Flowthrough.

pellet
denaturedRNA
chr. DNAProteins
cH₂O
PSD

⑥ washing buffer.

(ERP + CWC)

isopropanol + SDS)

⑦ Elution buffer.

nuclease free water (dissolve Na⁺)

Agarose gel electrophoresis

1% agarose gel

dissolve by microwave

1g agarose	→	100 ml TBE buffer
0.5g "	→	50 ml " "
1.5g "	→	150 ml " "
2g "	→	200 ml " "

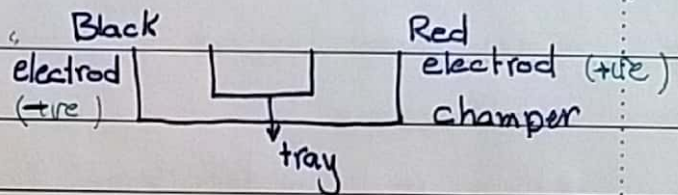
TBE(10x) stock

$$C_1 V_1 = C_2 V_2$$

$$10x \text{ } P = 1x \text{ } 1000 \text{ ml}$$

$$V_1 = 100 \text{ ml}$$

$$1000 \text{ ml } (10x) \rightarrow 900 \text{ ml (d H}_2\text{O)}$$



→ Ethidium Bromide

100 ml TBE → 1 ML ethidium bromide

50 ml TBE → 1 ML " "

150 ml TBE → 2 ML " "

200 ml TBE → 3 ML " "

Q.2)

Q.3) Gel electrophoresis

Technique used to separate macromolecules

NA

Proteins

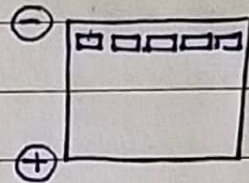
- | | |
|---------------------|---------------------|
| - Agarose | - Polyacrylamide |
| - charge, size (1D) | - charge, size (2D) |
| - Horizontal | - vertical |
| - low resolution | - more resolution |

Agarose gel electrophoresis

derived from seaweed.

porous material (sieve) full of Pores.

* Small Fragment of N.A move faster than large Fragment.



* to increase

* High resolution for large Fragment $\rightarrow$ large Pores size $\rightarrow$ low

* High resolution for small Fragment $\rightarrow$ small Pores size $\rightarrow$ high [agarose]

For genomics $\rightarrow$ 1%

For PCR Products $\rightarrow$ 2%

when we have to separate Fragmented N.A 40pb, 44pb $\rightarrow$ 3%
4%

TBE buffer

$\uparrow$ inactivation of DNAs

Tris-Borate EDTA $\rightarrow$ pH=8

tris-Base

Boric acid

$\rightarrow$ contain charge solute which carry current.

Ethidium Bromide (DNA stain)

(hydrophobic, aromatic)

alternative (Actinomycin D, Acridine orange)

- intercalating agent. (Red)

$\downarrow$
DNA $\rightarrow$ UV $\rightarrow$ orange.

* Sample loading

2 ML loading dye + 5 ML of sample (DNA) $\rightarrow$ Mix well.

loading dye:

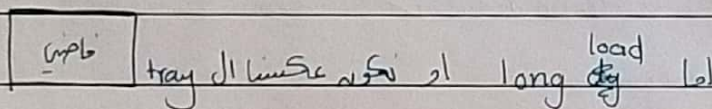
① Dense material (Force DNA to fall down into the well)

② Tracking dye (-ve charged, low M.W)

$\rightarrow$ visual monitoring of electrophoresis process

* DNA ladder

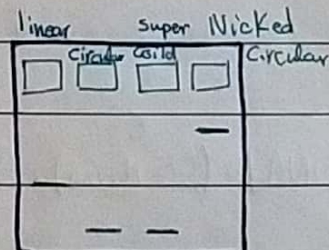
Mixture of DNA fragment ~~of~~ which known size)



DNA marker

710 —
260 —
230 —
200 —
150 —
100 —
50 —

the difference between the fragment
is not equal.

Shape with the same size

* Power supply

[agarose]	Voltage	time
1%	150 V	1/2 hr.
2%	120 V	1-1 1/2 hr
3%	100 V	2-2 1/2 hr.
4%	90 V	3-4 hr.

agarose di 1% w/v gel, 1% agarose

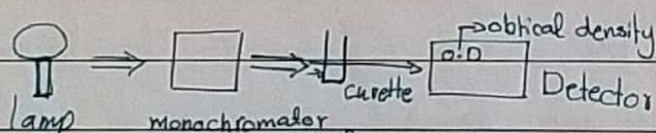
DNA analysis & Quantification:

- 1- Agarose gel electrophoresis. ✓ semi-quantitative method.
- 2- spectrophotometry
- 3- Nanodrop.

260nm ← N.A

Spectrophotometry

280 nm ← Proteins



↓ (split for spectra)
allow specific wave length
to transmit through it.

Beer's Lambert law.

$$C \propto A$$

$$C = E \times A$$

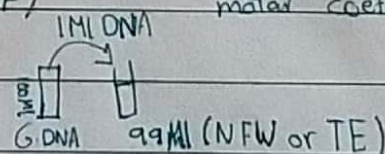
↙ molar coefficient.

$$[ds\ DNA] = (50\ \mu g/ml) \cdot A_{260nm} \times (D.F)$$

$$Dilution = \text{Previous dilution} \times \text{inoculum}$$

$$= \frac{1 \times 1\ \mu l}{100\ \mu l} = 10^{-2}$$

$$D.F = \frac{1}{D} = \frac{1}{10^{-2}} = 10^2$$



nitrogen base exposed more to the light

$$[ssDNA] = (33 \text{ Mg/ml}) A_{260} \times D.F \Rightarrow [dsDNA \approx 25\% \text{ absorbance}]$$

$$[ssRNA] = (40 \text{ Mg/ml}) A_{260} \times D.F$$

Purity

* $DNA = A_{260} / A_{280} \rightarrow [1.8-2]$ sample is Pure.

> 2 contamination with RNA

< 1.8 contamination with Protein

* $RNA = A_{260} / A_{280} \rightarrow [2.2-2.3]$ sample is Pure

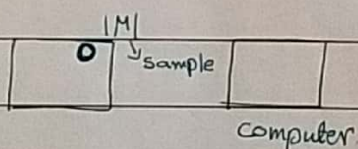
> 2.3 contamination with DNA

< 2.2 contamination with Protein

our sample contaminate with RNA we add RNase

" " " " Protein we go to our procedure
and to (نفسه) (نفسه)

Nano drop.



importance of gel electrophoresis to know if the sample is degraded or not.

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(PCR) Polymerase chain reaction

- technique ~~used~~ used to amplification of specific fragment of DNA → Kary Mullis have a nobel Prize in chemistry in 1993

$2(A+T) + 4(G+C)$
 (T_m) melting temperature (50% ds, 50% ss)
 value 60 → 55 (closed re. 60°C)
 Thermal cycler
 6. DNA
 5. Primer
 4. Primer
 3. Polymerase
 2. dNTPs
 1. buffer
 MgCl₂ (cofactor)
 Final PCR Product = 2^n , n: number of cycles.

Taq polymerase
Thermus aquaticus 72°C

3' GTACTGGATCGTACTAA GTACTTAGG TACT

Thermal cycler (PCR rxn) T.° Time # of cycle

1st cycle

Initial denaturation (90-97) (5-10m) 1

2nd cycle

1- denaturation (90-97) (30 sec)

(50-60°C) 2- Annealing $T_m - 5^\circ$ (30 sec)

3- extension (72°C) (30 sec)

3rd cycle

Final extension (72°C) (5-10m) 1

optional cycle

Hold 4°C ∞

Primers:

1- 100% complementarity

2- length 18-24 bp

3- GC% (40-60%)

4- T_m difference $< 5^\circ$

5- Complementarity $< 40\%$

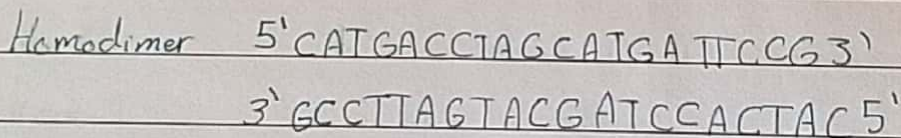
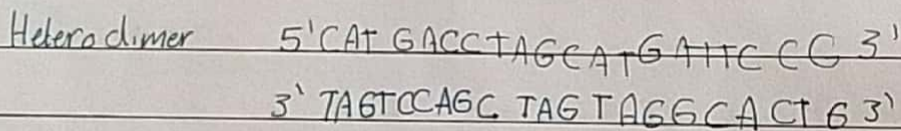
$$T_{m \text{ forward}} = 2(5+5) + 4(9+6) \\ = 20 + 40 = 60^{\circ}\text{C}$$

$$T_{m \text{ reverse}} = 2(5+5) + 4(6+5) \\ = 20 + 44 = 64^{\circ}\text{C}$$

Annealing $T_m - 5^{\circ}\text{C} = 55$
 انحراف 5 درجات

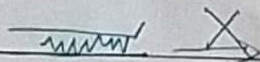
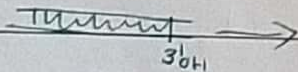
no specific
 annealing Temp. $\downarrow$

Primer dimer



self dimer (hair Pin) $\downarrow$ Primer الـ 18

non specific $\downarrow$ الـ 18



* Allele-specific PCR.

Master mix: taq^① polymerase, dNTPs^②, buffer^③, MgCl₂^④

PCR Product = 150 Pb

5'-ATCGATTAGGTACTAGTA^AGTACTAGTGTACTAGGTACTAGTACTACCTAGTG-3'
 5'-ATCGATTAGGTACTAGTA G T A G-3' 3'-ATGATCA T CAT GAT GGATCAC-5'

SNP: single nucleotide polymorphism.

Frequency $\geq 1\%$ Frequency of mutation $< 1\%$

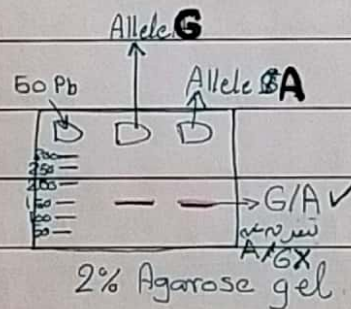
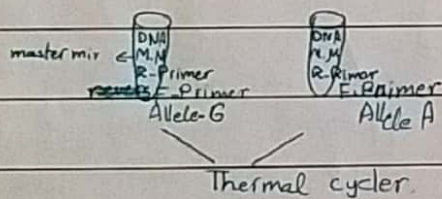
نقسم بـ ٢ أن SNP هو اختلاف بـ ١ نوكلوتا

Forward Primer: 5'-ATCGATTAGGTACTAGTA^A-3'

Forward Primer: 5'-ATCGATTAGGTACTAGTAG-3'

reverse Primer: 5'-CACTAGGTAGTAG^GACTAGTA-3'

One sample (2 PCR rxn)



— homozygous G

— homozygous A

— heterozygous A & G

VnTr: variable number of tandem repeat > 10 nt.

STR: Short Tandem repeat (2-6 nt)

Forensic

عدد التكرار الجيني في STR ATCG
 Primer $\uparrow$
 PCR-Product = 200 - (40) = 160 Pb

number of nucleotide repeat $\frac{160}{4} = 40$ repeat

عدد النوكليوتيدات المتكررة

Primers : Stock 1

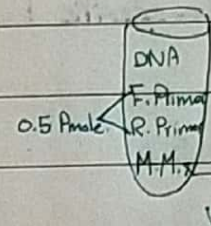
$$200 \text{ } \mu\text{M} = \frac{\text{no of moles (Picomoles)}}{\text{Volume (ML)}}$$

$$C_1 V_1 = C_2 V_2$$

$$10 \text{ } \mu\text{M} = 0.5 \times 25$$

$$V_1 = 1.25$$

$$= 1.3 \text{ ML}$$



In PCR IX

$$V = \frac{17.74 \times 10^3}{200} = 88.7 \text{ (NEW)}$$

$$C_1 V_1 = C_2 V_2$$

$$2 \times V_1 = 1 \times 25$$

$$V_1 = 12.5 \text{ ML}$$

stock 2 : working stock. 10 P.mole

$$C_1 V_1 = C_2 V_2$$

$$200 \cdot V_1 = 10 \text{ P.moles} \times 1000 \text{ ml}$$

PCR rxn one sample:

Master mix 12.5 ML

F Primer 1.3 ML

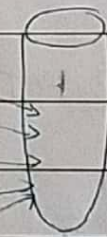
R. Primer 1.3 ML

NEW 7.9 ML

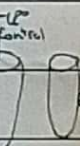
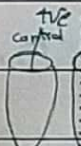
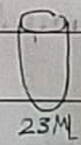
DNA 2 ML

25 ML

PCR rxn two sample



Vortex



23 ML

23 ML

23 ML

23 ML

2 ML

+2 ML

+2 ML

+2 ML

+2 ML

+2 ML

DNA1

DNA2

DNA

DNA

DNA

DNA

DNA

DNA

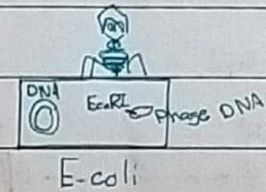
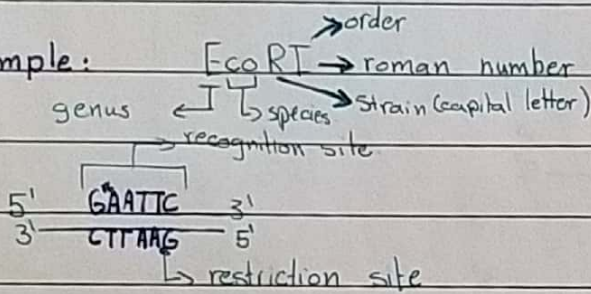
DNA

DNA

Restriction endonucleases enzyme.

- ① source of restriction enzyme is bacteria.
- ② Protein  DNA ^{sequence} at ~~specific site~~ (Recognition site) cut with it as specific site (restriction site)

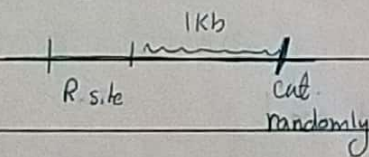
Example:



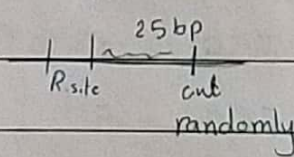
- ③ defense mechanism in Bacteria
- ④ methylation restriction modification system protect bacteria from methylase
- ⑤ recognition site is palindromic sequence (recognition site)

3-Types of restriction enzymes

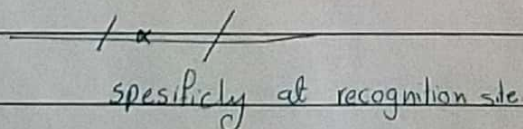
① Type I



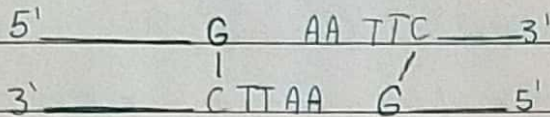
② Type III



③ Type II



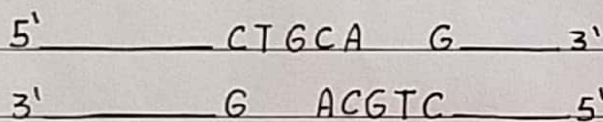
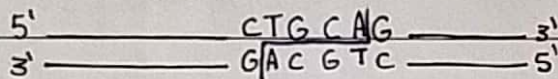
EcoRI



sticky end

5' over hang

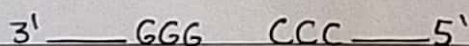
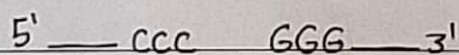
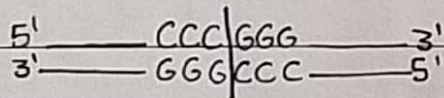
PstI



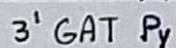
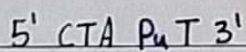
sticky end

3'-over hang

SmaI



Blunt end



Subject

Date

No.

5'-CCC GGG-3' SmaI (restriction enzyme)

RFLP: restriction Fragment length polymorphism

250bp

50bp

5'- ATGCCATGGTATAAIGTCAGCCCGGG AAT TTT CTA GTC ATT-3' (PCR product)

150

100

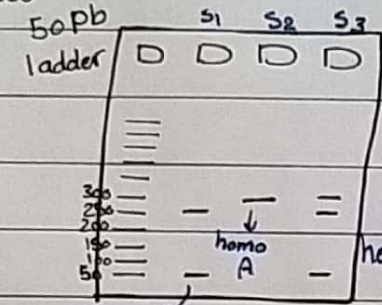
50bp

ladder

S1

S2

S3



cut C = 250, 50

not cut A = 300

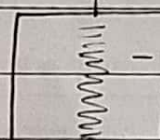
homozygous C 3%

Restriction enzyme Assay

0.1MI + 1m G.DNA

1/2hr - 37°C

enzyme OK ✓



DdeI

5'CTNAG3'

187bp

44bp

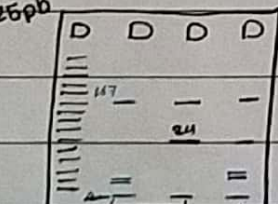
40bp

16bp

5' CTGATGTACGCTAAGATGCA TTAATCTAAGCGAAAAATCGTCGGACTAAGCCATGGC-3'

267bp

267bp



homo A

4%

homo C

hetero A/C

cut A → 187, 44, 40, 16

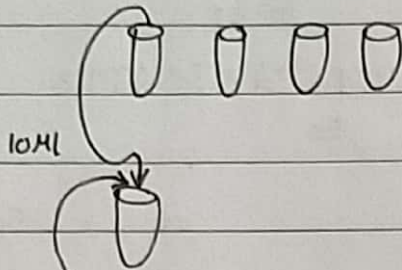
No cut C → 167, 84, 16

N O T E B O O K

RFLP rxn (1 sample)

20 μ L

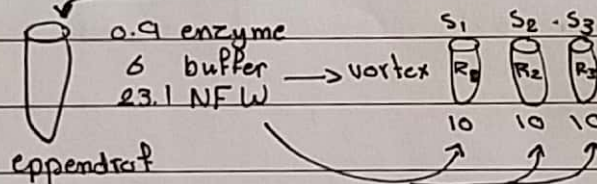
10 PCR product.



R. Enzyme (DdeI)	0.3 μ L
buffer	2 μ L
NEW	7.7 μ L
Final $\checkmark$	20

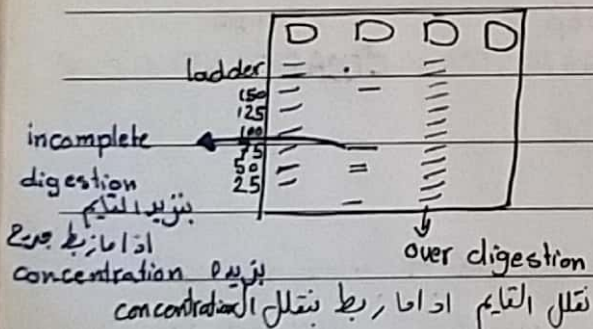
x3

RFLP (3 sample)

10 μ L
PCR
Product.

Incubation: 37°C

time: over night.



DNA sequencing

Genotyping $\left\{ \begin{array}{l} \text{allele specific PCR} \\ \text{RFLP} \\ \text{DNA sequencing} \end{array} \right.$

Point mutation or SNP

Gilbert, Sanger nobel prize 1980 in chemistry

* Gilbert method

(chemical degradation method) sequencing for short sequence

P
 $5'-ATTGACTTAGCC-3' \rightarrow$ ① Alkaline phosphatase: remove 5' Phosphate group
 ② Polynucleotide Kinase: add P^{32} at 5' end.

(X-ray)

DNA sequence + Dimethyl sulfate (G)	DNA sequence + Formic acid (AG)	DNA sequence + (hydrazine + NaCl) (C)	DNA sequence + hydrazine (C+T)	
Piperidine enzyme	Piperidine enzyme	Piperidine enzyme	Piperidine enzyme	
recognize modified nucleotide				
cut phosphodiester bond before modified nucleotide				
5'-ATT ^③	5'-ATT ^③	5'-ATTGA ^⑥	5'-A ^①	5'-ATTGACT ^⑦
5'-ATTGACTTA- ^⑤	5'-ATTG ^④	5'-ATTGACTTAG ^⑩	5'-AT ^②	5'-ATTGACTTAG ^⑧
	5'-ATTGACTT ^⑥	5'-ATTGACTTAGC ^⑪	5'-ATTGA ^⑤	5'-ATTGACTTAGC ^⑨
	5'-ATTGACTTA ^④	5'-ATTGAC ^④		

G	A & G	C	C & T	
		—	—	11
		—	—	10
—	—			9
				8
			—	7
			—	6
		—	—	5
	—			4
—	—			3
				2
			—	1

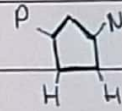
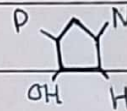
Polyacrylamide (X-ray)

5'-3'

5'-TTGACTTAGCC-3'

Sanger - method

(dideoxy terminators) enzymatic method



ATGACTACT

DNA Polymerase

deoxy
dNTPdideoxy
ddNTP

Primer

chain terminator

5'-GCATATGTCAGTCCA G-3'

^{P₃₂} 3'-CGTATACAGTCAGGTC-3'
 Primer 5'-CCAT

Heat

x Primer (labeled)

x DNA Polymerase

x dNTPs

Mix

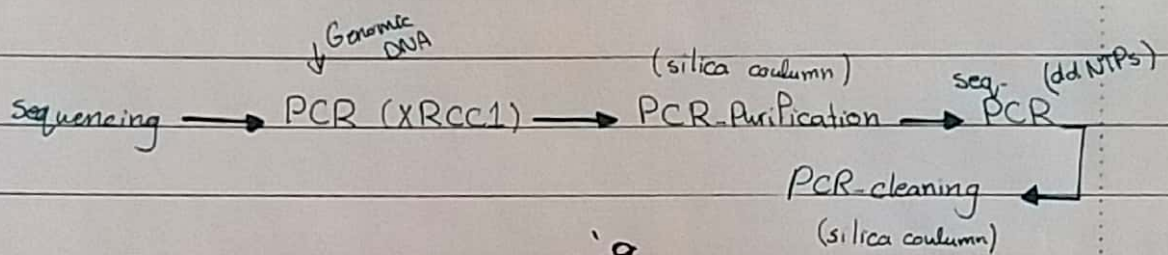
Primer ↑ 5'-n-A ①	Mix + ddATPs	Mix + ddTTPs	Mix + ddCTPs	Mix + ddGTPs
5'-n-ATGTCAG ②	5'-n-ATGT ④	5'-n-ATGTC ⑤	5'-n-ATG ⑥	
5'-n-ATGTCAG ②	5'-n-ATGT ④	5'-n-ATGTCAGTC ③	5'-n-ATGTCAG ⑦	
5'-n-ATGTCAGTCCA ⑩	5'-n-ATGTCAGT ⑨	5'-n-ATGTCAGTCC ⑪	5'-n-ATGTCAGTCCAG ⑧	

Mix + dd ATP	Mix + dd TTP	Mix + dd CTP	Mix + dd GTP
—			—
		—	
	—		
—		—	—
	—		—
			—
—	—		

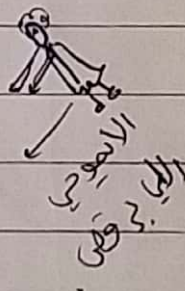
Polyacrylamide → X-ray

5'-3'

5'-ATGTCAGTCCAG-3'



في مرحلة



PCR

Master Mix 12.5 Ml

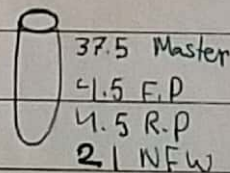
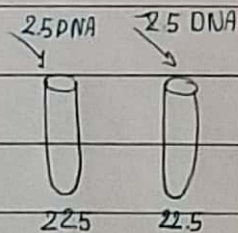
F. Primer 1.5 Ml

R. Primer 1.5 Ml

NEB 7

DNA 2.5

2.5



RT-PCR: Reverse transcriptase PCR

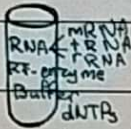
(وباء) (السرطان) (العدوى)

virus (RNA virus)

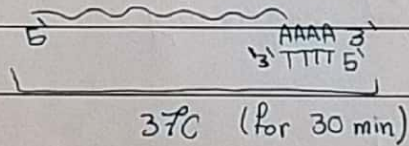
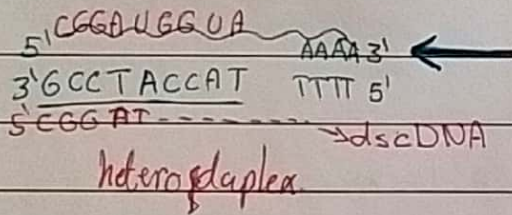
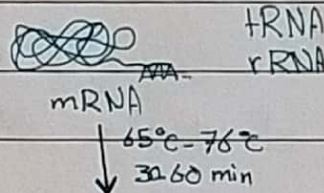
virus have enzyme (RT)
convert RNA to cDNA

⊗ tumor → viral infection
in animal cell

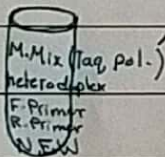
in the Past: RT-step.



reverse Primer ← Primer-oligo dT



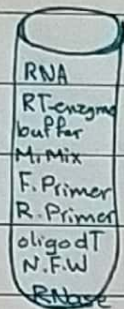
PCR step:



Thermal cycler - initial degradation (90°C)

Denaturation	95	30 sec	} 30 s
Annealing	50	30 sec	
extension	72	30 sec	
Final extension			

Nowadays ONE STEP

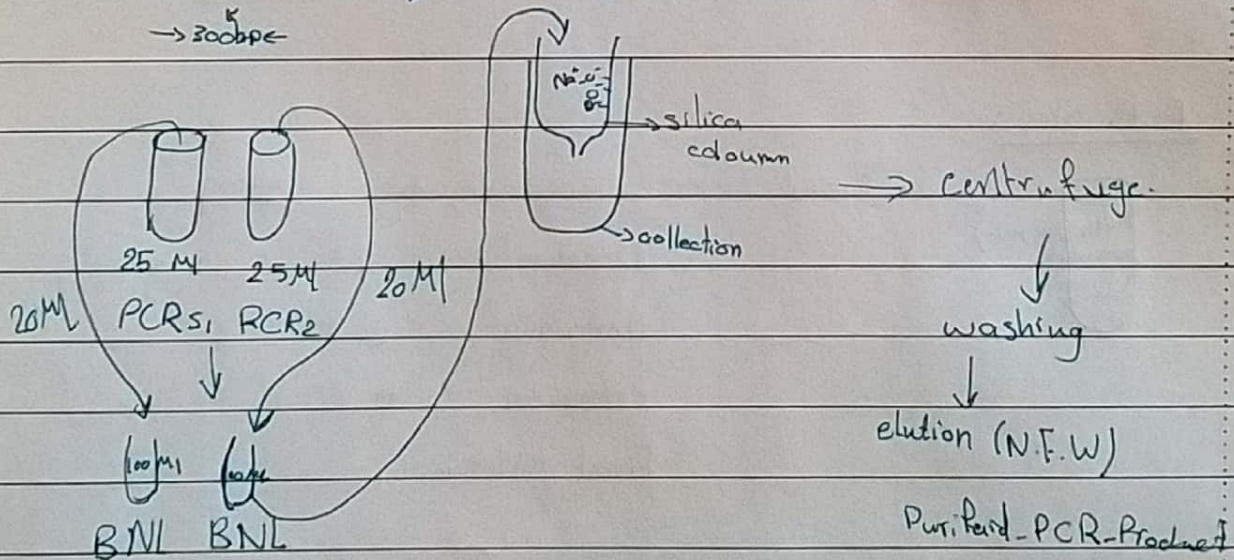


Thermal cycler

1- 65°C	30 min	1
2- 37°C	30 min	RNA → cDNA
3- 95°C	30 sec	denaturation → cDNA
4- Annealing	30 sec	} 30 times
5- 72°C	30 sec	
6- 72°C	5 min	
Final extension 7- 4°C	∞	dis. cDNA

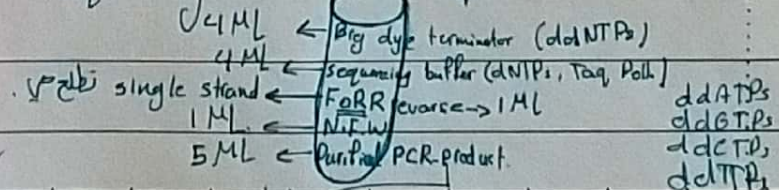
Automated sequencing

XRCC1 → PCR → PCR Purification

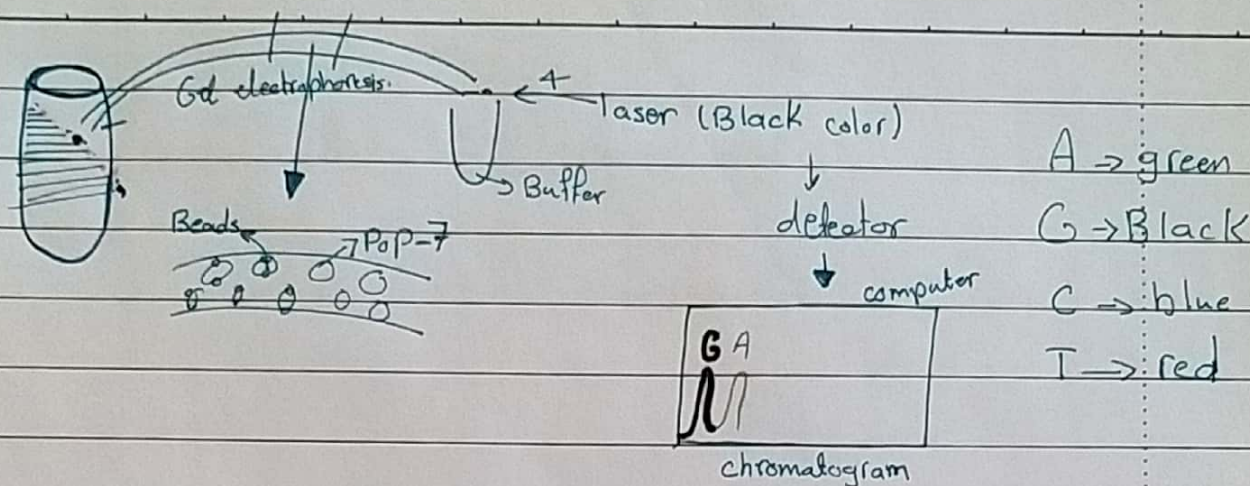


PCR Product → Thermal cycler → sequencing PCR

sequencing PCR



NOTEBOOK



NCBI → Blast (n) → Paste

xRec1

Blast: CGATAAGCCTAGACTAC

Blast

Blast (scan)

GG

AA

TT

CC

MM

Query: CGATAAGCCTAGACTAC

lab 1

lip 1

N

CIT